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COLOR IMAGE CODING USING REGIONAL CORRELATION OF PRIMARY COLORS

Yalon Roterman and Moshe Porat
Department of Electrical Engineering
Technion – Israel Institute of Technology
Haifa 32000, Israel

Abstract

Most color compression systems reduce the redundancies between the RGB color components by transforming the color primaries into a decorrelated color space, such as YIQ or YUV. In this paper a different compression approach is proposed. Since the high correlation of the RGB color channels implicitly suggests a localized functional relation between the components, it is used here in an alternative framework, by approximating subordinate colors as functions of a base color. This way, only a reduced number of parameters is required for coding the color information. Furthermore, since this correlation is in particular high locally, the image is first sub-divided into regions. For each region the correlation is analyzed and exploited separately. The size of the encoded regions is gradually reduced to allow progressively a more refined description of the transmitted image. Compression results of this progressive approach, which could be useful for slower communication channels, are presented, and compared with JPEG as a typical example of the decorrelation approach. Our conclusion is that the proposed new approach to progressive image coding could be superior to presently available compression techniques.

Corresponding author:
Moshe Porat,
Dept. of Electrical Engineering,
Technion, Haifa 32000, Israel.
Email: mp@ee.technion.ac.il
Fax: +972-4-829-5757
Tel.: +972-4-829-4684

1. Introduction

Natural scene images are characterized by high correlation between their R,G,B color components. Many compression techniques take advantage of this data redundancy and transform the colors primaries into a decorrelated color space such as YIQ (Y-luminance, I-inphase, Q-quadrature) or YUV. As the human visual system is more sensitive to details in luminance than to details in chrominance, the color information can be compressed at a higher rate. The JPEG compression algorithm is a typical example of this approach where the I and Q information of the YIQ representation are encoded separately at lower rates than the Y.

In recent years, however, a new approach has been developed, where instead of decorrelating the color components, the correlation is used to approximate two of the components as a function of the third component [1], [2], [3] and [4]. Furthermore, since color images also have spatial correlation, the information can be further compressed, describing an image as adjacent regions. An efficient approach that segments gray images into regions and achieves a high compression ratio is based on the Region Growing algorithm and was introduced in [5] and regarded as a 'second generation' compression method. In Region Growing we refer to the standard image processing technique that describes an image as a set of adjacent regions. The pixels inside the region represent the texture information of the image, and the pixels of the region's boundaries represent the contour details. In our proposed algorithm, contour and texture information are coded separately. We will show that the combined approach of the 'second generation' tools for image compression improves significantly the results reported in previous color compression algorithms.

2. The proposed algorithm

In this work we extend the principles of [1] and [5] to account for the two types of correlation: localized spatial correlation and the inter-color correlation. Combining the two methods – color coding using a color correlation technique and region segmentation technique for the base color can achieve a high compression ratio both for the base color and the color information. Furthermore, artifacts that result from the arbitrary 8x8 regions as in JPEG will be reduced when natural division in the image is used.

The stages in the proposed algorithm (Figure 2-1) are:



1. Construct the brightness image (Y) from the R, G and B components. In order to suppress false contours, the brightness image is filtered using the 'inverse gradient' filter [17]. This stage of preprocessing reduces the noise in the image while preserving the edges.
2. The filtered image is segmented into regions using the 'region growing' algorithm.
3. Artifacts and small regions are removed, as proposed in [5].
4. The contour information, of the segmented regions is coded using chain coding [18]. Since the direction of the contour tends to change only gradually, the differences in direction are encoded using a Huffman coding table.
5. For each region, the base color is approximated using a two-dimensional polynomial function.
6. The two subordinate colors in each region are encoded using a polynomial expansion of the base color

(2-1)
$$C_1 = a_k C_0^k + a_{k-1} C_0^{k-1} + \ldots + a_1 C_0 + a_0,$$
$$C_2 = b_k C_0^k + b_{k-1} C_0^{k-1} + \ldots + b_1 C_0 + b_0,$$

where C_0 is the base color, $C_{1,2}$ are interpolated subordinate colors, K is the order of the polynomial expansion and a_k, b_k are the coefficients of the polynomial expansion.

7. Finally, the coefficients of the color polynomial expansion and the polynomial coefficients of the base color in each region are quantized.

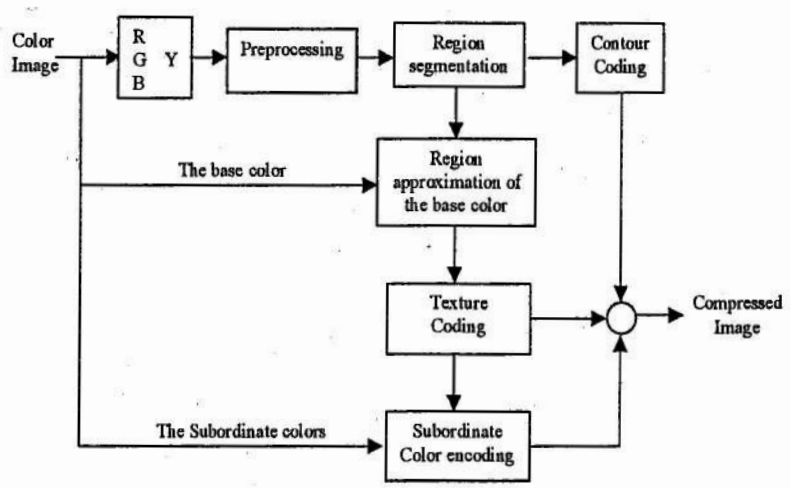


Figure 2-1 - The proposed encoder block diagram

The stages in the decoder (Figure 2-2) are carried out in a reverse order:

1. From the compressed data, the contours are reconstructed. Then each region is approximated using the polynomial coefficients of the regions. The result of this stage is the segmented base color.
2. The subordinate colors are reconstructed using the color coefficients and the reconstructed base color.
3. Each contour is approximated as the average of the texture surrounding it.

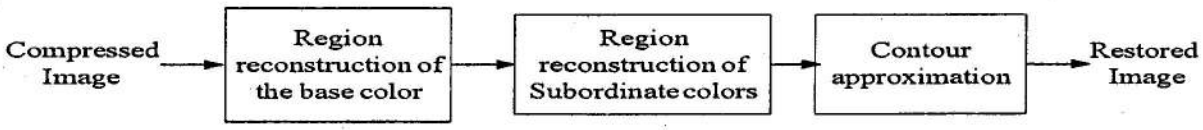


Figure 2-2 - The proposed decoder block diagram.

2.1 Preprocessing

Since the human eye is more sensitive to the luminance than to the chrominance information, the segmentation will be done according to the luminance. The first stage is to construct the luminance information from the RGB components. This Y component is likely to best represent edges in the image, regardless of their actual color, R, G, or B.

If the segmentation process is done on the original image, the noise that appears as small variation of the gray level will produce undesired contours. The preprocessing stage prepares the image for the segmentation process by reducing the noise in the image. In this work the Inverse Gradient Filter [17] is used due to its two main advantages:

- The inverse gradient filter reduces the variance of each region while preserving the average.
- The inverse gradient filter has little effect on the sharpness of edges in the picture.

Image smoothing algorithms are usually done by a mask. The center pixel is replaced with the result of the operator. Many masks are composed of fixed coefficients, the coefficients can be all equal or decreasing when the distance to the center is getting larger. The result of applying this kind of mask on the image is smoothing and noise reduction. The down side to this process is that the edges of the image are also smoothed and the filtered imaged is blurred. If we notice that the changes of the gray level inside a

region are smaller than the changes between regions, we can use a mask with coefficients that are inversely proportional to the gradient.

If we use a mask composed of coefficients that are inversely proportional to the gradient, as suggested in [17], the contours will contribute little to the filtering while a neighboring pixel that has a close gray level will affect more on smoothing the image.

2.2 Segmentation

In the second stage of the algorithm, the image is segmented into regions. The pixels that are inside a region represent the texture information and the pixels that surround the regions represent the contour information.

The segmentation starts with a seed pixel and examines the properties of neighbouring pixels. If the neighboring pixels have similar properties they are added to the region. The procedure stops when no additional pixels satisfy the criteria to be added to the region, or when the algorithm reaches the image boundary. This procedure partitions the image into regions in which the brightness is continuous and smooth.

The property that is used in this work to determine if neighboring pixels belong to the same region is of a fixed gray-level interval. Although it has a constant width, this interval is made adaptive by moving it up and down on the gray-level scale in order to intercept maximum number of pixels. This displacement is constrained, however, so that previously intercepted pixels always remain in the region.



(a) (b)

Figure 2-3 - The segmented Lena image after the region growing process. Segmentation results of the (a) original image, and (b) image after applying the inverse gradient filter with 10 iterations are shown for gray_level_interval=10.

In Figure 2-3 we can see that the inverse gradient filter reduces the number of false contours and the number of regions while preserving the edges. The choice of a gray level interval influences the results of this algorithm. If this value is too small, a large number of regions is obtained, and the compression ratio will decrease. If this value is too large, the number of regions is small, leading to a high compression ratio but with poor image quality.

After the segmentation process there are two types of artifacts: contours that do not completely separate a region and contours that are two pixels wide, such as in the figure below.

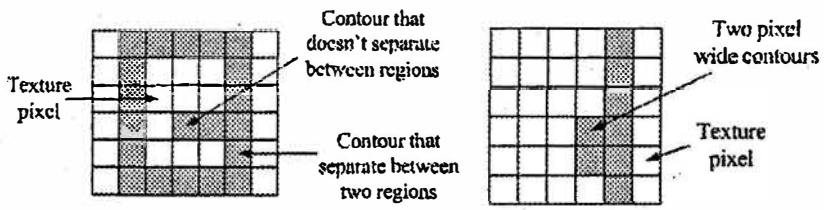


Figure 2-4 – artifact contours

The new-segmented image after removing those artifacts is shown in the figure below.



Figure 2-5 - The segmented Lena image after removing artifact contours

When comparing Figure 2-3 with Figure 2-5, we can see that the number of contours was reduced (from 17865 to 14427) and the number of regions was not changed.

- From here on, the results that will be presented are after applying this preprocessing on the images.

2.3 Reducing the number of regions

At this stage, the resulting image can be viewed as that of a puzzle with one-point thick closed contours. Unfortunately, the number of these contours could be too high to achieve efficient compression. Two heuristics are used to decrease the number of regions obtained by region growing: elimination of small regions and merging weakly contrasted adjacent regions [5].

2.3.1 Elimination of small regions

The histogram and the cumulative histogram of the number of regions as a function of their size indicate that roughly 70 percent of the regions have less than 15 pixels. There are two reasons for this result: First, the granularity is not completely eliminated during preprocessing, and second, small areas of high gradient in the original image are segmented into several small regions. If it is assumed that regions containing a number of pixels less than a threshold are not significant. Their elimination decreases drastically the number of regions. To avoid creation of holes in the image, these regions are added to one of their adjacent regions. To minimize the corresponding distortion, the enclosing region is chosen as the adjacent region whose mean gray level is closest to that of the small region to be added.



Figure 2-6 - The segmented Lena image after removing small regions. The above process was applied to Figure 2-5, where regions smaller than 10 pixels were removed.

This process, applied to Figure 2-5 reduced the number of regions to 792 (reduction of about 70%) and the number of contours to 11328.

2.3.2 Removing close contrasted regions

Observing areas of constant luminance gradient in the original image and the segmented image in Figure 2-6, it can be noticed that some regions are subdivided into sub regions although there is no real contour, for example, the area above Lena's hat and some areas in Lena's face. This is due to the mechanism of region growing which divides the image into regions of fixed gray-level dynamic range.

The second approach to decrease the number of regions is thus to merge adjacent regions whose contrast is below a certain level. The region will be merged into an adjacent region whose contrast is the smallest. The contrast between adjacent regions is defined as the mean of the absolute gray level difference calculated along their common border.

The result of applying this process to the image of Figure 2-6 is shown in Figure 2-7.



Figure 2-7 - The segmented Lena image after merging close contrasted regions (closer than 10)

After this procedure, the number of contours is 6991 and the number of regions is 300.

2.4 Contour coding

Efficient description of the contours obtained after the segmentation procedure should be transmitted as part of the coded information. A precise description of contours is essential to the human visual system.

There are several methods to code contour information. In [5] the information is approximated as lines and arcs. This work uses the Freeman chain code method with 8 connected contours. However, since certain pairs of codes are much more likely to occur in a chain we employ a statistical approach. We assume that lines turn slowly, and sharp bends are unusual in natural images, i.e.

- In most cases the next movement is likely to be the same direction as the previous one,
- Next most likely movement is 45 degrees right or left, and
- 180 degrees is the least likely next movement.

This calls for Huffman coding to improve the compression ratio. Instead of transmitting the direction of the next contour, the change in the direction will be transmitted.

Furthermore, the starting point of each chain requires a large number of bits, according to the size of the image. Here to save bits too, only the distance between the new starting point and the previous one is transmitted. The maximum distance between two starting points of two successive chains will determine the number of bits needed to code each distance. This number will be transmitted in the header of the information.

2.5 Base color approximation

The purpose of this stage is to provide a simple description to each region in the base color, usually the green. After the segmentation process, there are no sharp discontinuities in the gray level within a region; this implies that a region can be described by a smooth two-dimensional function. This approximation is obtained using a polynomial function. The order of the polynomial is a trade off between the approximation error and the cost involved in coding the polynomial coefficients. The coefficients of each polynomial are calculated so that the MSE is minimized.

The region's coefficients are transmitted in a 'scan' manner – the order of the regions is from left to right. This way, when the decoder had reconstructed the region matrix (from the contours information) it knows which region the coefficients belong to.

In order to decode the region information, the encoder must transmit the order of each polynomial (0, 1 or 2) and the coefficients of each polynomial. And since most regions are approximated by a zero order polynomial function, zero order is coded with one bit (0) whereas first and second order will be coded with two bits (01 and 10 respectively).

Result of applying the proposed algorithm and the JPEG algorithm to a gray image can be seen in Figure 2-8 and Figure 2-9.



(a) (b)

Figure 2-8 - Compressed gray image of Lena with the proposed algorithm where the gray level interval is (a) 10, and (b) 40. The compression ratios are 20 and 40, with PSNR of 24.66db and 24.46 db, respectively.



Figure 2-9 - Compressed gray image of Lana with the JPEG algorithm.

The compression ratio here is 36 with PSNR of 24.47db.

Table 2-1 - Compression result of gray image of Lena

Gray level interval	Number of zero order regions	Number of first order regions	Number of second order regions	Number of contours	Compression ratio
10	213	40	47	6991	20
40	90	6	6	4767	40

Results:

- The quality of the image compressed by the region growing is better than the image compressed by the JPEG algorithm.
- Increasing the gray level parameter increases the compression ratio.
- Most regions are described by a zero order polynomial function.
- Increasing the compression ratio in the region-growing algorithm by removing additional small region does not have a substantial affect on the image quality. On the other hand, increasing the compression ratio in the JPEG algorithm will cause the algorithm to transmit only DC parameters and will decrease the image quality substantially.
- The number of regions in the image after region growing is around 100, as apposed to 1024 in JPEG. If we wish to code the subordinate colors as a function of the base color, the number of the color coefficients is much smaller than by using the region growing technique.

2.6 Subordinate color encoding

In order to utilize the color correlation between the RGB components, each region in the subordinate colors (R, B) is approximated as a function of the region in the base color (G). The approximation is done using a first order polynomial function according to (2-2).

(2-2)
$$B = b_1 \cdot G + b_0$$
$$R = r_1 \cdot G + r_0$$

The coefficients of the polynomial function that minimize the MSE are calculated as follows:

$$b_0 = \frac{1}{n}\sum_1^n B_i \qquad r_0 = \frac{1}{n}\sum_1^n R_i$$

(2-3)
$$b_1 = \frac{\sum_1^n B_i \cdot G_i - \frac{1}{n}(\sum_1^n B_i)\cdot(\sum_1^n G_i)}{\sum_1^n B_i^2 - \frac{1}{n}(\sum_1^n B_i)^2} \qquad r_1 = \frac{\sum_1^n R_i \cdot G_i - \frac{1}{n}(\sum_1^n R_i)\cdot(\sum_1^n G_i)}{\sum_1^n R_i^2 - \frac{1}{n}(\sum_1^n R_i)^2}$$

where n is the number of pixels in the approximated region and r_1, b_1, r_0, b_0 are the coefficients of the polynomial approximation.

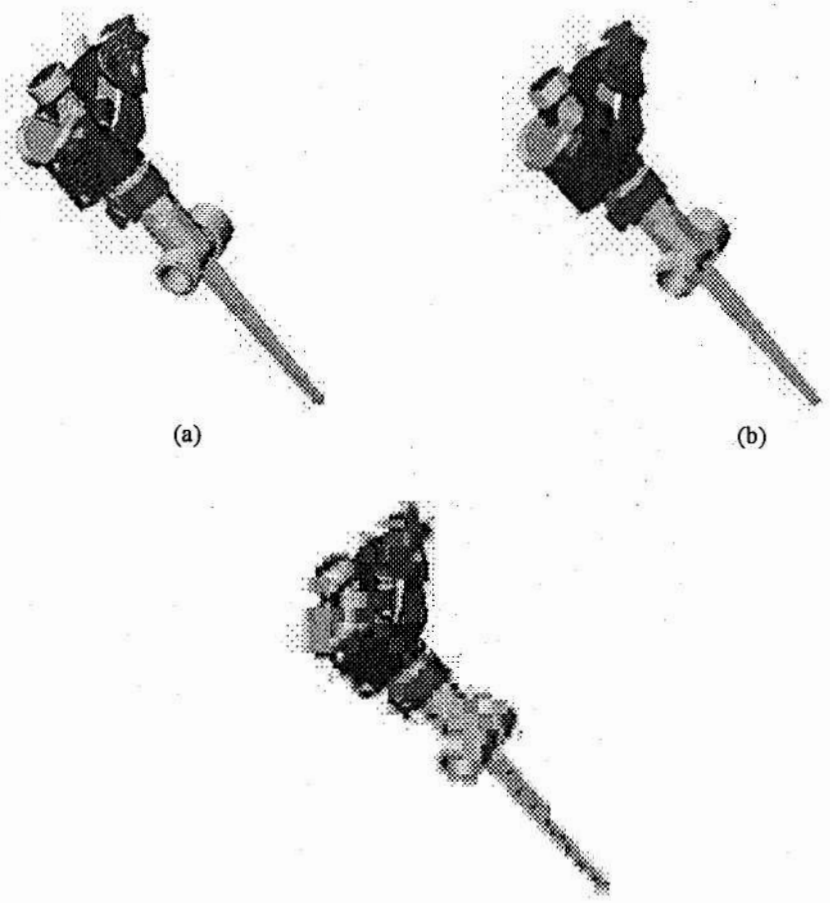
For each region, 4 coefficients have to be transmitted: 2 for each subordinate color. Since most regions of the base color are described by a zero order polynomial function, r_1 and b_1 cannot be calculated. The solution to that problem can be found in the fact that the order of the polynomial function that describes the base color is transmitted. If the base color is constant the two other colors will also be approximated as constant. That means that only r_0 and b_0 will be transmitted causing the compression ratio to increase further.

2.7 Quantization

The quantization process is carried out for the polynomial coefficients of segmented regions and for the polynomial coefficients of the color approximation. In this work, these coefficients are uniformly quantized and values below or above a global threshold limits are clamped. Here the threshold is a function of the mean and standard deviation. We have noticed that quantization of several coefficients together leads to poor results. For that reason each color coefficient or region coefficient is quantized separately.

3. Compression results

Results of applying the proposed algorithm to several images are shown in Figures 3-1 and 3-2.



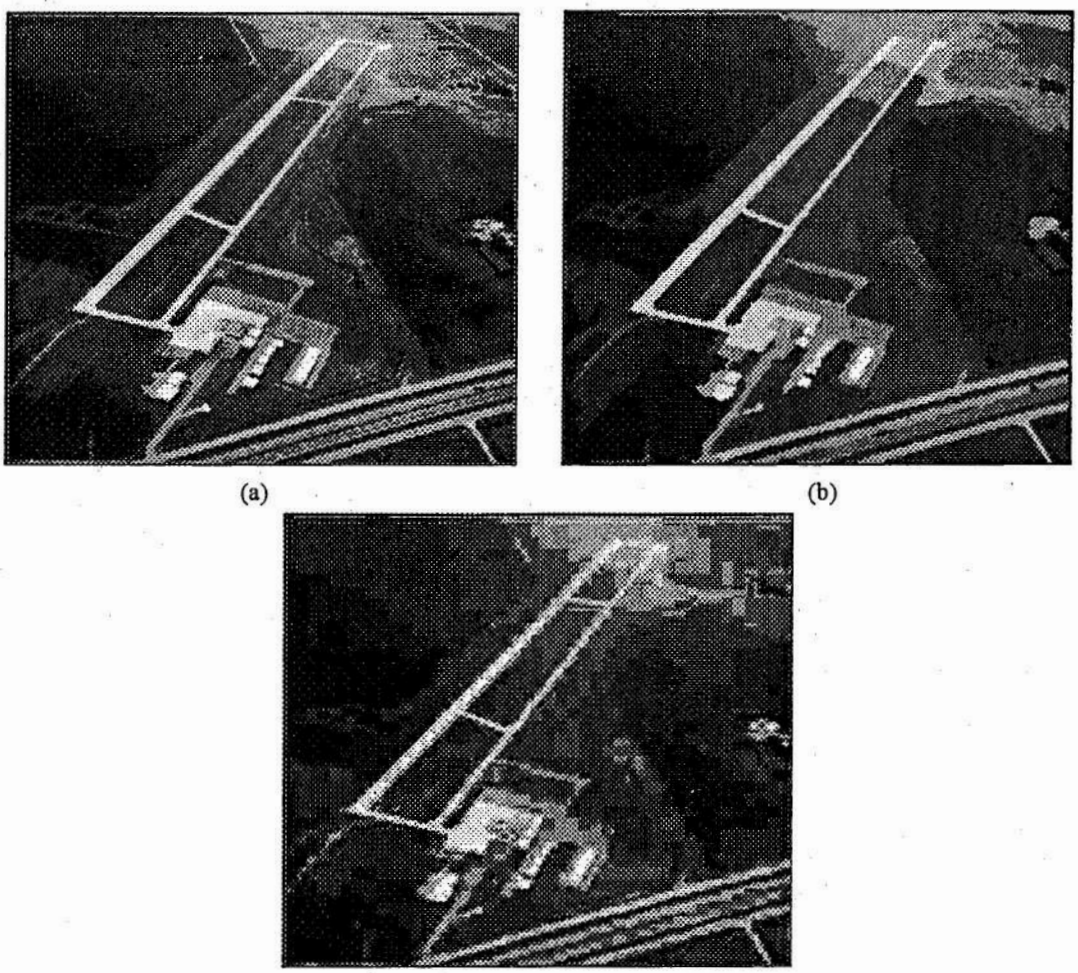
(a) (b)

(c)

Figure 3-1 – Compression results for solid works picture. (a) The original image, image compressed by (b) the proposed algorithm and (c) the JPEG algorithm with compression ratio of 243 and 95 and PSNR of 29.3db and 25db respectively.

It can be seen that the compression ratio and the PSNR are much higher for the proposed algorithm when compared with JPEG. The proposed algorithm naturally

works most efficiently for images with well-defined regions, like aerial photos as in the figure below:



(a) (b)

(c)

Figure 3-2 – Compression results for aerial picture (a) The original image, image compressed by (b) the proposed algorithm and (c) the JPEG algorithm with compression ratio of 86 and 83 and PSNR of 22.3db and 22.2db respectively.

As mentioned earlier, increasing the compression ratio for the JPEG algorithm leads to a significant degradation in image quality, whereas the proposed algorithm is more robust since only additional small regions are removed.

Furthermore, since the compression ratio of the JPEG algorithm is bounded, the advantages of the proposed algorithm become more apparent in images with fewer regions like in Figure 3-1. Another observation is that applying the proposed algorithm to

the same image but with larger dimensions results in better compression results. The reason for this is that the number of regions remains almost the same, but the size of each region is larger. In the JPEG algorithm this is not the case since the regions are always 8x8.

The algorithm was implemented on a Pentium 4, 2.4 GHz computer with a 256 MB RAM. It was written and tested using MATLAB. The average number of operations for typical images was 290M – 250M operations for the encoding stage and approximately 40M operations for decoding. For this MATLAB-based system, the encoding-decoding average time was 2.5 minutes. For hardware implementation, the execution time is expected to be in the order of a second, assuming that a DSP-based system can execute approximately 250M operations per second (or more, in case of parallel processing).
Cellular Networks are a natural application of this algorithm. Today cellular bandwidth is growing, however, for a typical size image of 1-2 Mbits compressed by a factor of 50-100, the time to transmit the image is also in the order of a second. These numbers indicate that the compression and reconstruction time will be in the order of the transmission time and will not have a significant effect on the user.

4. Progressive transmission

To improve the efficiency of the transmission, in particular for slower channels, the information can be sent progressively. In this process we reduce the size of the regions gradually. We start with transmission of large regions, and then sub-divide the regions to improve the resolution of the reconstructed image. The first compressed image contains only large regions, and then in each consecutive stage, new and smaller regions are added to produce a more refined image.

All the stages carried out according to this approach are the same as the basic algorithm, except the stage of reducing the number of regions (Section 2.3). Instead of reducing the number of regions in one step, the reduction is done in a several steps according to a defined parameter. In each step larger regions are removed and finally the close contrasted regions are removed. In each stage the region matrix will therefore have fewer and larger regions in each step. An example of progressive transmission can be seen in the figure below.

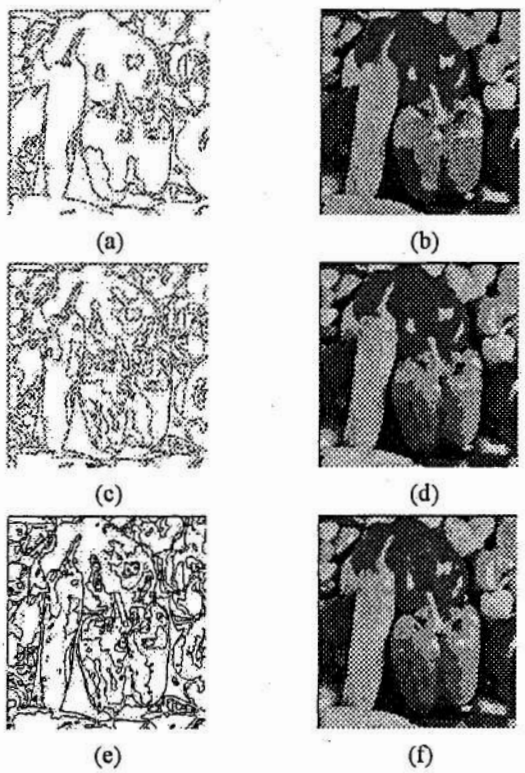


Figure 4-1 - Region matrix and reconstructed image in progressive transmission (a), (b) first stage, (c), (d) second stage and (e), (f) third stage.

The decoder in this case reverses this process. First the decoder reconstructs the contours from the transmitted data, then finds the region that should be updated and accordingly reconstructs the base color and the subordinate colors. Note that the statistics of the contour's chains is different in each stage. The length of the chains decreases in later steps, so the code for end-of-chain appears more frequently. This implies that each stage may need a different Huffman table. The selected code was the one that minimizes the bit per direction. Results of progressive transmission can be seen in Figures 4-2, 4-3 and 4-4.

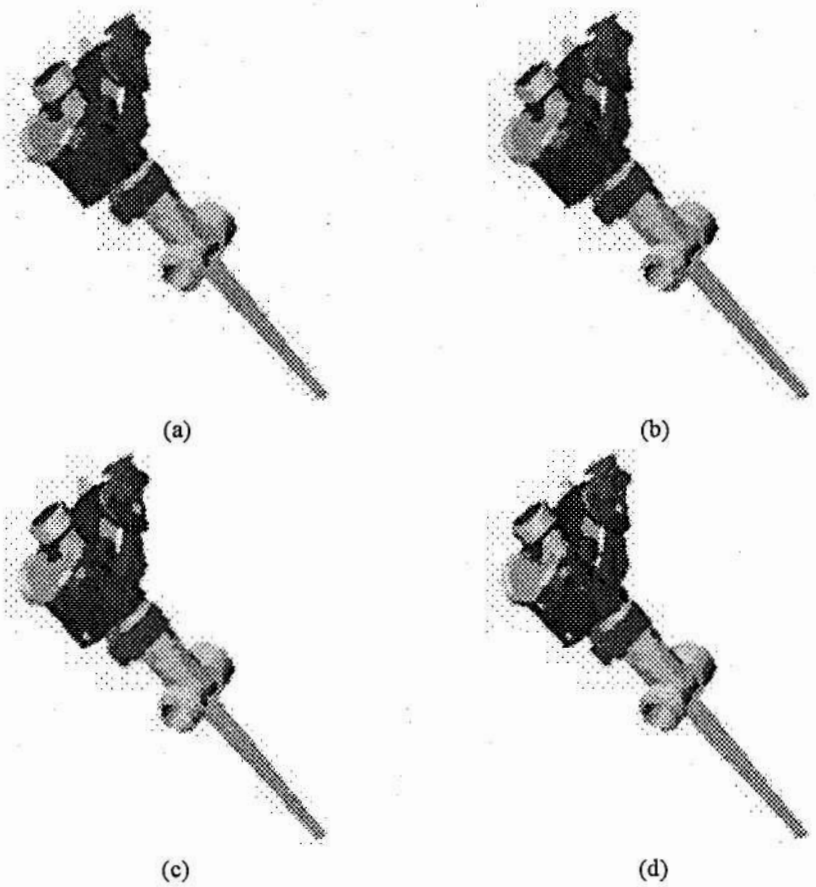


Figure 4-2 - Stages in progressive transmission for the solid works image (a) The first transmitted image, (b), (c), (d) are the results of the next stages of the progressive transmission with a compression ratio of 230, 179, 110 and 61 respectively.

The compression ratio, for each stage, takes into account the bits that were transmitted in previous stages and the bits that are transmitted in the current stage.

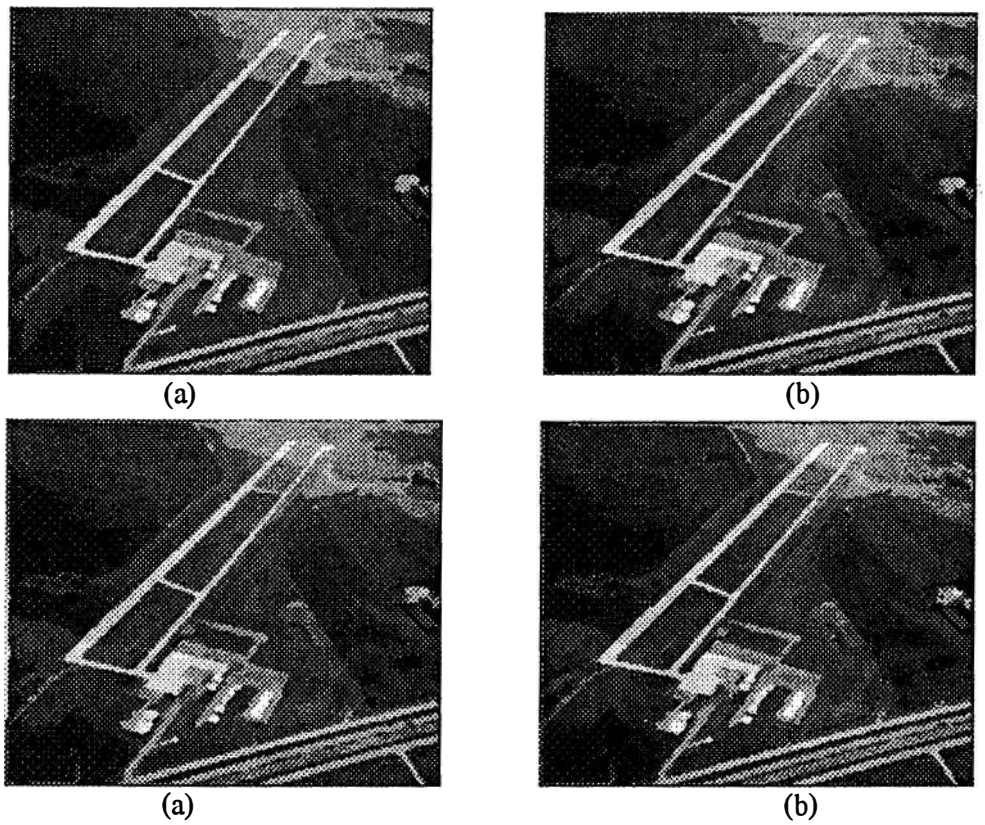


Figure 4-3 - Stages in progressive transmission for the aerial image. (a) The first transmitted image, (b), (c), (d) are the results of the next stages of the progressive transmission with a compression ratio of 83, 43.1, 33.4 and 21.8 respectively.

Generally, the JPEG algorithm achieves better results than the proposed algorithm for low compression ratios. The break-even point depends on the image. For complex images like Lena, the break-even point is a compression ratio of approximately 65. For images with well-defined region, the break-even point is approximately 40. However, if we use progressive transmission and the receiver requires a higher quality image it may be more efficient to transmit only new smaller region than to transmit the entire JPEG image.



Figure 4-4 - Stages in progressive transmission for the Lana image. (a) The first transmitted image, (b), (c), (d) are the results of the next stages of the progressive transmission with a compression ratio of 82.2, 50, 37 and 22.7 respectively.

5. Summary and conclusions

In this work we have combined for the first time two major principles of image compression: intercolor correlation and contour-texture image representation to achieve an efficient compression method. Unlike most compression techniques, which reduce the correlation between the RGB components by transforming the RGB into a decorrelated color space, in this work a new compression approach is introduced exploiting the correlation between the primary colors.

Furthermore, since this correlation is in particular high locally, the image is first sub-divided into regions, using the region-growing algorithm. This way the image is

described according to the natural regions and the blockiness effect that appears in other compression techniques is not apparent. We have demonstrated that an algorithm that uses the intercolor correlation and the spatial correlation achieves a high compression ratio with higher image quality compared to the JPEG algorithm.

To summarize, this work has led to several conclusions:

- The results indicate a compression ratio of 80-230 for typical images, with PSNR of above 25db.
- These results are superior to the JPEG algorithm, especially for images with well-defined regions, such as aerial mapping or solid works drawings. In these images the effects of JPEG are very noticeable. Unlike these arbitrary 8x8 blocks of JPEG, the regions in the proposed algorithm are based on natural segmentation of the image, providing an efficient tool for higher quality content-based compression.
- The image quality and compression ratio can be controlled by several parameters and can be adjusted accordingly.
- In JPEG, for a high compression ratio, further increase causes a significant degradation in image quality. Increasing the compression ratio in the proposed algorithm is less apparent since only small regions are removed.
- Since the compression ratio of the JPEG algorithm is bounded, the proposed algorithm has more advantage in images with fewer regions as in Figure 3-1.
- Generally, it can be shown that larger images achieve a higher compression ratio.

We have also introduced an extension to this algorithm that allows progressive transmission. In this process we reduce the size of the regions gradually. We start with the transmission of large regions, and then sub-divide the regions to improve the resolution of the reconstructed image in each stage. Compression results of this progressive approach, which could be useful for slower communication channels, have been presented.

Our conclusion is that a regional color correlation approach could be superior to the traditional decorrelation methods.

Acknowledgments

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